GHULAM ARBI

Postdoctoral Researcher — Computer Vision & Deep Learning

arbi@mail.ustc.edu.cn | +86-1307-553-2010 | Hefei, China (relocation-ready)
ORCID: 0009-0002-7365-0293 | GitHub: github.com/aiguo112 | LinkedIn: linkedin.com/in/arbiarbii



RESEARCH OBJECTIVE

Seeking a postdoctoral research position in computer vision / deep learning, with a focus on panoramic (360°) vision, remote sensing image segmentation, or geometry-aware scene understanding. Available to relocate internationally immediately following doctoral degree conferral.

PROFILE

PhD in Computer Vision, University of Science and Technology of China (USTC), with 10+ years of combined programming and software engineering experience and 15 papers (10 accepted, 5 under review). Designs and evaluates deep learning algorithms for segmentation, detection, counting, classification, and forecasting across 360° panoramic, remote sensing, SAR-optical, space-target, medical, and autonomous-driving imagery. Specializes in near-zero-overhead lightweight attention modules, rigorous baselines, and careful evaluation.

EDUCATION

University of Science and Technology of China (USTC) *2018 – 2026*
PhD, Computer Science and Technology

Thesis: "Exploring Segmentation in Panoramic Vision Using Attention Mechanisms"

Advisor: Prof. Yanyong Zhang — Executive Dean, School of Artificial Intelligence and Data Science, USTC

South China University of Technology (SCUT) *2016 – 2018*
MS, Electrical and Computer Engineering

Thesis: "Research on Fast R-CNN for Visual Object Detection"

Advisor: Prof. Yu Zhuliang — Executive Dean, Wu Xianming School of Intelligent Engineering, SCUT

Islamia University of Bahawalpur, Pakistan *2010 – 2014*
BS, Computer Science

Final project: Android city-navigation app for Bahawalpur with local-business tagging, in the spirit of Lonely Planet's local discovery guides

RESEARCH EXPERIENCE

PhD Researcher, Computer Vision — USTC *2018 – Present*

- Researched deep learning algorithms for 360° ERP panoramic imagery, spanning semantic segmentation, crowd counting, scene classification, and object detection.
- Designed lightweight distortion-aware attention modules (latitude-aware sinusoidal weighting, SE/CBAM channel recalibration, circular padding) adaptable to CNN and Transformer backbones with near-zero parameter overhead.
- Built and publicly released three annotated benchmark datasets advancing panoramic computer vision research.
- Co-supervised undergraduate and master's students; authored teaching materials on deep learning and attention mechanisms.

Independent Research — Vision-Only 3D Occupancy Prediction *2026 – Present*

- Implemented an LSS-based BEV occupancy baseline from scratch in PyTorch on nuScenes/Occ3D (34k samples).
- Vectorized BEV pooling for a 3.2× speedup with bit-exact results; temporal ego-pose-aligned fusion raised mIoU from 0.128 to 0.147 in pilot experiments.
- Introduced UniOcc per-voxel optical-flow supervision and quantitatively verified ego/object motion decomposition (0.0000 m/s residual).

OmniOrbit: Panoramic Space-Situational-Awareness Benchmark — First Author *2026*

- Generated 49,998 synthetic ERP panoramic images using Blender and NASA 3D assets (13 object classes, 4 tasks).
- Quantified a ≈63-point same-frame projection gap (ERP 90.4% vs. perspective 27.4% top-1); ERP retains 100% of targets vs. ≈70% loss for a fixed perspective camera.
- Synthetic-to-real transfer on real NASA imagery reached 49.2% ± 4.5% linear-probe accuracy.

OmniSplit360: Edge–Cloud Split Computing for Panoramic Segmentation — First Author *2026*

- Profiled 3,990 configurations of SegFormer-B0 and modeled per-tile edge–cloud allocation as a Multiple-Choice Knapsack Problem (MCKP), solved exactly via dynamic programming.

- Real two-device TCP experiments: at a 12 ms budget, compute-only accounting admits 266/266 tiles; accounting for measured network latency admits 0/266, exposing a critical cost-model gap.

MS Researcher, Computer Vision — SCUT *2016 – 2018*

- Researched Fast R-CNN-based visual object detection; designed CNN architectures; ran GPU-accelerated training with PyTorch and TensorFlow.

PRIOR PROFESSIONAL EXPERIENCE

Assistant Software Engineer, ISETech *Apr 2015 – Jun 2016*

- Developed and maintained Android mobile applications using the Android SDK; diagnosed and resolved production defects.

Assistant Software Engineer, VexxorTech *Aug 2014 – Jan 2015*

- Contributed to Android application development, including feature implementation and bug fixes.

Android Developer, Clone Logic *Jan – Dec 2013*

- Developed and maintained Android applications, covering UI implementation and application logic.

PUBLICATIONS (15: 10 ACCEPTED, 5 UNDER REVIEW)

Accepted

1. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "Beyond Perspective: Latitude-Aware Crowd Counting in Equirectangular Panoramic Imagery." **PRICAI 2026**
2. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "Text-Steered Modality Routing for Annotation-Free Open-Vocabulary SAR-Optical Segmentation." **ICAIMS 2026**
3. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "SA-RSRefSeg: Scale-Aware Attention Prompting for Referring Remote Sensing Image Segmentation." **PRCV 2026**
4. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "Detect-Then-Segment: A Unified Pipeline for Non-Cooperative Space Target Perception." **PRCV 2026**
5. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "LatentPano: Latitude-Aware Spatial Transformer for 360° Panoramic Scene Classification." **IICAIET 2026**
6. **Ghulam Arbi**. "Content-Conditioned Cyclic Seam Carving for Equirectangular Panoramas." **AMICT 2026**
7. Talal Ahmad, Husnain Abid, and **Ghulam Arbi**. "Leakage, Shortcuts, and Shift: An Evaluation-Integrity Audit of Smartphone-Based Anemia Screening." **AMICT 2026**
8. Talal Ahmad, Husnain Abid, and **Ghulam Arbi**. "HemaFormer: Report-Supervised Multi-Task Learning for Anemia and RBC Morphology on AneRBC." **AMICT 2026**
9. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "The Cost Model Is the Bottleneck: Per-Tile Split Allocation for 360° Segmentation with Measured Network Latency." **IEEE CSCloud 2026**
10. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "OmniOrbit: Advancing Space Situational Awareness with a Multi-Task 360° Vision Benchmark." **ACM Multimedia Asia 2026**

Under Review

11. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "PanoDive360: A Novel Equirectangular Projection Dataset and Enhanced Attention Mechanism for Underwater 360-Degree Multiclass Semantic Segmentation." **ACM Multimedia Asia (MMAsia)**
12. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "ERP-SegFormer: Feature Recalibration for Outdoor Panoramic Image Segmentation with Squeeze-and-Excitation Block." **ACM Multimedia Asia (MMAsia)**
13. **Ghulam Arbi**, Naveed Ur Rehman Junejo, and Abid Hussain. "Phenological Contrast: A Geography-Agnostic Prior for Cross-Geography Field Boundary Segmentation." **IEEE JSTARS**
14. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "Tinyship-Fingerprint: Compact Open-Set Individual Vessel Identification from Underwater Acoustic Emissions." **ICASSP 2027**
15. **Ghulam Arbi**, Lu Zhang, and Yanyong Zhang. "When Contextual Evidence Conflicts: Causal Localization of Order-Sensitive Evidence Selection in LLMs." **ECIR 2027**

HONORS & FUNDING

Chinese Government Scholarship (CSC) — Master's and PhD studies

10 accepted papers, 5 under review; 3 publicly released panoramic-vision benchmark datasets

TECHNICAL SKILLS

Programming:

Python, Java, C++, C#, SQL, Bash

Deep Learning: PyTorch, TensorFlow, Hugging Face, CUDA/AMP, TensorRT, ONNX, model quantization

Computer Vision: Segmentation, detection, 360°/ERP imagery, depth estimation, attention/SE mechanisms, density-based counting, occupancy prediction

Architectures: Transformer, SegFormer, DeepLabv3+, U-Net, Swin/ViT, DETR, Mask R-CNN, YOLOv8

Deployment: TensorRT, ONNX, model quantization, CUDA/AMP, edge–cloud split inference

Tools: OpenCV, Blender, CARLA, Docker, Git/Linux, pandas, scikit-learn, Weights & Biases, LaTeX

LANGUAGES

English — Professional working proficiency | Chinese (Mandarin) — HSK Level 3 | Urdu — Native

REFERENCES

Prof. Yanyong Zhang — PhD Advisor, Executive Dean, School of Artificial Intelligence and Data Science, USTC

Email: [to be confirmed]

Additional referees available upon request.